

Complete Probability *Library*

Chapters 1–5 · Every concept, every edge case, every exam scenario — deeply explained

Set Theory · Probability · Conditional Probability · Bayes · Random Variables · Discrete Distributions

Pedagogy: Hook → Mechanics → Contrast → Trap → Worked Examples → Aha Moment

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CHAPTER 01 Set Theory

01 HOOK

What is a set?

Think of a set as a labelled bag. You put objects into it. Two bags are equal only if they contain *exactly* the same objects — order doesn't matter and duplicates don't count. The sample space S is the master bag: every possible outcome lives inside it, and every event is a smaller bag inside S .

A set must be **well-defined**: for every possible object, it must be decidable whether it belongs or not.

02 MECHANICS

Notation & Definitions

DEFINITION 1.1.1

Set — Two Ways to Write One

Roster (listing): $A = \{1, 2, 3\}$ — list every element.

Set-builder rule: $B = \{x \mid 0 \leq x \leq 5, x \in \mathbb{Z}\} = \{0, 1, 2, 3, 4, 5\}$

Interval (continuous): $C = [1, 4]$ means all real x with $1 \leq x \leq 4$. Note: $C \neq \{1, 2, 3, 4\}$!

Reserved symbols: $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$, $\mathbb{R}$ = all reals.

DEF 1.1.1.2

Universal / Sample Space

S = the set of all possible outcomes of a random experiment

Every event $A \subseteq S$. The sample space is your universe — nothing outside it exists in the model.

DEF 1.1.1.3

Empty Set $\emptyset$

$\emptyset = \{\}$ — contains no elements

$\emptyset \subset A$ for every set A . In probability: $P(\emptyset) = 0$. The impossible event.

DEF 1.1.1.4

Subset $A \subset B$

$A \subset B \Leftrightarrow (\forall x)(x \in A \Rightarrow x \in B)$

Every element of A is also in B . $A \not\subset B$ means at least one element of A is NOT in B .

DEF 1.1.1.5

Equal Sets

$A = B \Leftrightarrow A \subset B$ and $B \subset A$

The standard proof method: show each is a subset of the other.

DEF 1.1.1.6

Power Set $\mathcal{P}(A)$

$\mathcal{P}(A) = \{\text{all subsets of } A\}$, $|\mathcal{P}(A)| = 2^{|A|}$

If $A = \{1, 2, 3\}$:

$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}$ — 8 subsets = 2^3 .

DEF 1.2.1

Intersection $A \cap B$

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Disjoint: $A \cap B = \emptyset$ — mutually exclusive. Generalised: $\bigcap_{i=1}^n A_i$.

DEF 1.2.3

Union $A \cup B$

$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

"Or" is inclusive. $A \subset A \cup B$ always. Generalised: $\bigcup_{i=1}^n A_i$.

DEF 1.2.4

Complement $A^c = \bar{A}$

$A^c = \{x \mid x \in S, x \notin A\}$

$A \cup A^c = S \quad A \cap A^c = \emptyset \quad (A^c)^c = A$

DEF 1.2.5

Set Difference $A - B$

$$A - B = \{x \mid x \in A, x \notin B\} = A \cap B^c$$

NOT commutative: $A - B \neq B - A$ in general.

$$A - B = A \text{ iff } A \cap B = \emptyset.$$

CRITICAL DISTINCTION — $\in$ VS $\subset$

$\in$ relates an *element* to a set: $3 \in \{1, 2, 3\}$. $\subset$ relates a *set* to a set: $\{2, 3\} \subset \{1, 2, 3\}$. Never write $\{2, 4\} \in \{1, 2, 3, 4\}$ — that set $\{2, 4\}$ is not listed as a member. Also: $\emptyset \subset A$ for every set (vacuously true), but $\emptyset \in A$ only if $\emptyset$ is explicitly an element.

MECHANICS

Set Operations — Laws & Identities

THEOREM 1.4.3 — MEMORISE THESE

De Morgan's Laws

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \qquad \overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$\text{General: } \overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \bar{A}_i \qquad \overline{\bigcap_{i=1}^n A_i} = \bigcup_{i=1}^n \bar{A}_i$$

Memory trick: "Break the bar, change the operator." $\cup$ flips to $\cap$ and vice versa.

THEOREM 1.4.1

Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Proof sketch (1st law): $x \in A \cap (B \cup C) \Rightarrow x \in A$ and $(x \in B \text{ or } x \in C) \Rightarrow (x \in A \cap B) \text{ or } (x \in A \cap C) \Rightarrow x \in (A \cap B) \cup (A \cap C)$. Reverse inclusion symmetric. $\square$

PROPERTY	STATEMENT	KEY USE
Commutativity	$A \cup B = B \cup A, \quad A \cap B = B \cap A$	Order doesn't matter
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$	Grouping doesn't matter
Idempotency	$A \cup A = A \cap A = A$	Repeated ops harmless

PROPERTY	STATEMENT	KEY USE
Identity	$A \cup \emptyset = A, \quad A \cap S = A$	$\emptyset$ neutral for $\cup$; S neutral for $\cap$
Annihilation	$A \cap \emptyset = \emptyset, \quad A \cup S = S$	S absorbs under $\cup$
Complement	$A \cup \bar{A} = S, \quad A \cap \bar{A} = \emptyset, \quad \overline{\bar{A}} = A$	Double negation cancels
Monotone subset	$A \cap B \subset A \subset A \cup B$	Intersection shrinks; union grows
Difference identity	$A - B = A \cap \bar{B}$	Convert difference to intersection for proofs
Chain 1	$(A - B) - C = A - (B \cup C)$	De Morgan + associativity
Chain 2	$A - (B \cup C) = (A - B) \cap (A - C)$	Distributing difference over union

⚠ CONTRAST

 $\subset$ vs $\in$ — and discrete vs continuous sets

The two most confusing distinctions in set theory:

(a) $\subset$ vs $\in$: $D = \{2, 4\}$ and $C = \{1, 2, 3, 4\}$. Write $D \subset C \checkmark$. Never $D \in C$ — $\{2, 4\}$ is not listed as a member of C , only the numbers 2 and 4 are.

(b) Discrete vs continuous: $C = \{1, 2, 3, 4\}$ (four specific integers) $\neq E = \{x \mid 1 \leq x \leq 4\} = [1, 4]$ (uncountably many reals). When someone asks "is $C = E$?", check immediately whether one is an integer list and the other an interval.

(c) Strict vs non-strict boundaries: Finding $P \cap Q$ where $P = \{x \mid 0 \leq x \leq 10\}$ and $Q = \{x \mid 5 < x < 15\}$. Always track $<$ vs $\leq$ at every boundary.

⚠ THE TRAP

 De Morgan direction — the most common error

Why it feels correct: Students write $\overline{A \cup B} = \bar{A} \cup \bar{B}$, thinking "the complement just passes inside." It looks symmetric.

Why it is wrong: $\overline{A \cup B} = \bar{A} \cap \bar{B}$ — the operator FLIPS. $\bar{A} \cup \bar{B}$ is actually $\overline{A \cap B}$, a completely different expression. Confusing these will produce wrong answers on every set theory proof.

Memory: "Break the bar $\rightarrow$ change the operator." If you see $\cup$ inside, the complement gives $\cap$ outside, and vice versa.

WORKED EXAMPLES

Algebraic proof technique

Example A: Show $(A - B) - C = A - (B \cup C)$.

- 1 Convert difference: $(A - B) - C = (A \cap \bar{B}) \cap \bar{C}$
- 2 Associativity: $= A \cap (\bar{B} \cap \bar{C})$
- 3 De Morgan (reverse): $\bar{B} \cap \bar{C} = \overline{B \cup C}$
- 4 Result: $A \cap \overline{B \cup C} = A - (B \cup C) \quad \square$

Example B: Show $A \subset B \Leftrightarrow A = A \cap B$.

- 1 $(\Rightarrow)$ If $A \subset B$: take any $x \in A$. Then $x \in B$ (by subset), so $x \in A \cap B$. Thus $A \subset A \cap B$. But $A \cap B \subset A$ always. So $A = A \cap B$.
- 2 $(\Leftarrow)$ If $A = A \cap B$: then $A = A \cap B \subset B$. So $A \subset B$. $\square$

Example C: Let $U = \{1, \dots, 10\}$, $A = \{1, \dots, 8\}$, $B = \{2, 3, 4\}$, $C = \{1, 3, 5, 7, 9\}$. Find $A \cap (\bar{B} \cup C)$.

- 1 $\bar{B} = U - B = \{1, 5, 6, 7, 8, 9, 10\}$
- 2 $\bar{B} \cup C = \{1, 5, 6, 7, 8, 9, 10\} \cup \{1, 3, 5, 7, 9\} = \{1, 3, 5, 6, 7, 8, 9, 10\}$
- 3 $A \cap (\bar{B} \cup C) = \{1, \dots, 8\} \cap \{1, 3, 5, 6, 7, 8, 9, 10\} = \{1, 3, 5, 6, 7, 8\}$

AHA!

The empty set is a subset of everything

$\emptyset \subset A$ for every set A , including $\emptyset \subset \emptyset$. The proof is vacuous truth: "every element of $\emptyset$ belongs to A " is true because $\emptyset$ has no elements — there is nothing to falsify the claim.

Consequence: A set with n elements has exactly 2^n subsets (including $\emptyset$ and itself). A set with 3 elements has 8 subsets.

Also notable: the empty set is BOTH a subset of every set AND an element of some sets — $\emptyset \in \{\emptyset, 1, 2\}$. Context determines which.

CHAPTER 02 Probability Concepts

0 HOOK

Probability as a score on uncertainty

Imagine scoring every event: impossible events score 0, certain events score 1, and everything else lives strictly in between. The entire field of probability is just a rigorous system that keeps this scoring consistent. The three Kolmogorov axioms are the only rules needed — all other results follow logically from them.

0 MECHANICS

Axioms, Theorems, Counting

DEFINITION — SAMPLE SPACE, EVENTS, PROBABILITY

S = sample space = set of all possible outcomes. An **event** is any subset $A \subseteq S$. An **elementary event** $\{a\}$ contains exactly one outcome. **Mutually exclusive events:** $A \cap B = \emptyset$.

Empirical / frequency definition: $P(A) = \lim_{n \rightarrow \infty} \frac{r}{n}$ where r = times A occurs in n trials.

KOLMOGOROV AXIOMS – EVERY PROBABILITY RESULT FOLLOWS FROM THESE THREE

Axioms of Probability

AI. $0 \leq P(A) \leq 1$ for all $A \subseteq S$

AII. $P(S) = 1$

AIII. $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$

General (pairwise mutually exclusive): $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$

THEOREM 2.2.1

Complement Rule

$$P(\bar{A}) = 1 - P(A)$$

Often easier to compute the complement then subtract.

THEOREM 2.2.2

Partition Rule

$$P(A) = P(A \cap B) + P(A \cap \bar{B})$$

Splits A into two disjoint pieces using any event B.

THEOREM 2.2.3

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Subtract the intersection once to avoid double-counting.

When mutually exclusive, $P(A \cap B) = 0$.

THEOREM 2.2.4

Monotonicity

$$A \subset B \Rightarrow P(A) \leq P(B)$$

Use this as a sanity check before any calculation.

CLASSICAL MODEL – EQUALLY LIKELY OUTCOMES

Classical Probability Formula

$$P(A) = \frac{n(A)}{N}$$

$n(A)$ = favourable outcomes, $N = n(S)$ = total equally likely outcomes. **Requires** all outcomes equally likely.

Permutations & Combinations

Permutations of n distinct objects: $n!$

$$\left(\left\{_{n}P_r = \frac{n!}{(n-r)!}\right\} \quad \text{order matters, } r \text{ from } n\right)$$

Repetition allowed: n^r arrangements of r positions each with n choices

$$\binom{n}{r} = \frac{n!}{(n-r)!r!} \quad (\text{order does not matter})$$

$$\text{Multinomial: } \binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}, \quad \sum n_i = n$$

Key question: Does the order matter? YES $\rightarrow$ permutation. NO $\rightarrow$ combination.

CONTRAST**Permutation vs Combination — order is everything**

The single deciding question: Does swapping two selected items produce a different outcome?

- **YES $\rightarrow$ Permutation ${}_nP_r$:** PIN codes, race placements (1st/2nd/3rd are different), arrangements of letters.
- **NO $\rightarrow$ Combination $\binom{n}{r}$:** Committees (president-secretary-treasurer is ALSO a combination if the roles are unspecified), lottery numbers, card hands.

Note: $\binom{n}{r} = {}_nP_r / r!$. Dividing by $r!$ removes all orderings of the chosen r items.

Multinomial counts ways to divide n distinct items into k groups of sizes $n_1, \dots, n_k$. Used for tutorial group assignments, bridge hands dealt to multiple players.

THE TRAP**Three probability impossibilities to check immediately**

Before any calculation, verify these bounds:

- $P(A \cap B) \leq \min(P(A), P(B))$ — intersection can't be larger than either component.
- $P(A \cup B) \geq \max(P(A), P(B))$ — union can't be smaller than either component.
- $P(A \cup B) \leq P(A) + P(B)$ — union can't exceed the sum.

Example: $P(A) = 0.2$, $P(B) = 0.5$, $P(A \cap B) = 0.25$. Stop immediately — $0.25 > P(A) = 0.2$. **Impossible.**

Example: $P(A) = 0.8$, $P(B) = 0.4$, $P(A \cup B) = 0.7$. Stop — $0.7 < \max(0.8, 0.4) = 0.8$. **Impossible.**

Example A — Cellphone number (no repeats, starts with 7, 2nd digit $\neq 0$):

- 1 Digit 1 = 7 (fixed): 1 choice.
- 2 Digit 2: cannot be 0 or 7 (already used). Choices: $10 - 2 = 8$.
- 3 Digits 3–8: no further restriction beyond no repeats. From 8 remaining: $8 \times 7 \times 6 \times 5 \times 4 \times 3$.
- 4 Favourable: $1 \times 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3$.
- 5 Total (only constraint: start with 7, no repeats): $1 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3$.
- 6 $P = 8/9 \approx 0.889$.

Example B — Two brothers, tutorial groups of 25/30/20 from 75 students:

- 1 Total ways: $\binom{75}{25,30,20} = 75!/(25! 30! 20!)$.
- 2 Fix both brothers in same group — three cases.
Both in group of 25: choose 23 more from 73: $\binom{73}{23,30,20}$.
Both in group of 30: $\binom{73}{25,28,20}$. Both in group of 20: $\binom{73}{25,30,18}$.
- 3 Simplify using the shortcut: $P = \frac{n_1(n_1 - 1) + n_2(n_2 - 1) + n_3(n_3 - 1)}{N(N - 1)}$ where $N = 75$.
- 4 $P = \frac{25 \cdot 24 + 30 \cdot 29 + 20 \cdot 19}{75 \cdot 74} = \frac{600 + 870 + 380}{5550} = \frac{1850}{5550} = \frac{1}{3} \approx 0.333$.

Impossible combinations reveal hidden constraints

Checking the monotonicity bounds can tell you an answer is impossible before you compute anything. More powerfully: it lets you work backwards. If you know $P(A) = 0.6$, $P(B | A) = 0.3$, and $P(A \cup B) = 0.72$, you can recover $P(B)$:

$$P(A \cap B) = P(B | A) \cdot P(A) = 0.18. \text{ Then}$$

$$P(B) = P(A \cup B) - P(A) + P(A \cap B) = 0.72 - 0.6 + 0.18 = 0.30.$$

Then independence check: $P(A)P(B) = 0.6 \times 0.30 = 0.18 = P(A \cap B)$. **They are independent.**

HOOK

Restricting the sample space

Analogy: You roll a die. Someone tells you "the result is even." You instantly discard 1, 3, 5 — your universe shrinks from 6 outcomes to 3. Conditional probability is exactly this: every piece of information restricts the universe to only the outcomes compatible with that information.

Formally, $P(B | A)$ is the probability of B *within the restricted universe* where A is true. The denominator $P(A)$ rescales everything so the restricted probabilities sum to 1.

MECHANICS — CONDITIONAL PROBABILITY

DEFINITION 3.1.1 — THE FOUNDATIONAL FORMULA

Conditional Probability

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0$$

Plain English: "What fraction of A 's probability mass is also in B ?" Satisfies all three Kolmogorov axioms — conditional probability is a valid probability measure on the reduced sample space.

KEY PROPERTY

Since $P(B | A) + P(\bar{B} | A) = 1$, you can use the complement rule within conditionals: if it's easier to find $P(\bar{B} | A)$, then $P(B | A) = 1 - P(\bar{B} | A)$.

Also: $P(A \cap B) = P(B | A) \cdot P(A) = P(A | B) \cdot P(B)$. This is the **Multiplication Rule**, and it is used constantly.

MULTIPLICATION RULE – GENERAL CHAIN FORM

Chain Rule

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_2, A_1) \dots P(A_n | A_{n-1}, \dots, A_1)$$

Each factor conditions on all previous events. For $n = 2$: $P(A \cap B) = P(A) P(B | A)$. For $n = 3$:

$$P(A \cap B \cap C) = P(A) P(B | A) P(C | A, B).$$

DEFINITION 3.3.1 – ONE OF THE MOST IMPORTANT DEFINITIONS IN ALL OF STATISTICS

Statistical Independence

$$A \text{ and } B \text{ independent} \Leftrightarrow P(A \cap B) = P(A) P(B)$$

$$\text{Equivalently: } P(B | A) = P(B) \text{ or } P(A | B) = P(A)$$

Meaning: knowing A occurred gives zero information about B .

Closure under complementation: If $A \perp B$, then also $A \perp \bar{B}$, $\bar{A} \perp B$, $\bar{A} \perp \bar{B}$.

CHECKING INDEPENDENCE – THE THREE EQUIVALENT TESTS

All three are equivalent: (1) $P(A \cap B) = P(A)P(B)$, (2) $P(B | A) = P(B)$, (3) $P(A | B) = P(A)$. In practice, test (1) is easiest since it only requires knowing the joint and marginal probabilities.

ⓘ CONTRAST

Independence vs Mutual Exclusivity — opposites, not synonyms

PROPERTY	MUTUAL EXCLUSIVITY	INDEPENDENCE
Definition	$P(A \cap B) = 0$	$P(A \cap B) = P(A)P(B)$
Meaning	They CANNOT both occur	One gives NO info about the other
Implication	Knowing A occurred RULES OUT B — maximum dependence	Knowing A changes nothing about B
When compatible?	Always possible for any two events	Requires $P(A), P(B) > 0$ for non-trivial cases
Both simultaneously?	Only if $P(A) = 0$ or $P(B) = 0$ (trivial)	Two non-trivial mutually exclusive events are DEPENDENT

The key paradox: Two non-trivial mutually exclusive events are as *dependent* as possible — if one happens, the other definitely didn't. Independence and mutual exclusivity are near-opposites for events with positive probability.

⚠ THE TRAP

Confusing $P(A | B)$ with $P(B | A)$

This is the single most costly error in conditional probability. $P(A | B) \neq P(B | A)$ in general — these are different quantities.

- $P(\text{test positive} | \text{has disease}) = 0.98 \neq P(\text{has disease} | \text{test positive}) \approx 0.30$
- $P(\text{defective} | \text{Machine A}) = 0.10 \neq P(\text{Machine A} | \text{defective}) \approx 0.625$

This is the **Prosecutor's Fallacy** in law: confusing $P(\text{evidence} | \text{innocent})$ with $P(\text{innocent} | \text{evidence})$.

Rule: Identify the given event (after the "|") and the event you want (before the "|"). If the question gives you the result and asks for the cause — that is Bayes' Theorem.

📘 WORKED EXAMPLES

Example — Independence test: $P(A) = 0.6$, $P(B | A) = 0.3$, $P(A \cup B) = 0.72$. Find $P(B)$ and classify.

- 1 $P(A \cap B) = P(B | A) \cdot P(A) = 0.3 \times 0.6 = 0.18$
- 2 From addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow P(B) = 0.72 - 0.6 + 0.18 = 0.30$
- 3 Independence test: $P(A)P(B) = 0.6 \times 0.30 = 0.18 = P(A \cap B)$. ✓ **Independent.**
- 4 Mutual exclusivity: $P(A \cap B) = 0.18 \neq 0$. **NOT mutually exclusive.**

Total Probability & Bayes' Theorem — Deep Treatment

HOOK

Working backwards from effect to cause

Standard probability: We know the cause, and we ask about the effect. Machine A has a 2% defect rate → what is $P(\text{defective} | \text{from Machine A})$? Easy: it's given.

Bayes' theorem: We observe the effect, and we ask which cause produced it. We find a defective item → what is $P(\text{from Machine A} | \text{defective})$? This requires Bayes.

The entire idea: *"We see the effect. Which cause most likely produced it?"* Bayes is the mathematically correct way to reverse a conditional probability.

PREREQUISITE – REQUIRED BEFORE APPLYING BAYES

Law of Total Probability

$$P(B) = \sum_{i=1}^n P(B | A_i) P(A_i)$$

Requirements for $\{A_i\}$: (1) Pairwise mutually exclusive: $A_i \cap A_j = \emptyset$ for $i \neq j$. (2) Exhaustive: $\bigcup_i A_i = S$, equivalently $\sum_i P(A_i) = 1$.

WHY TOTAL PROBABILITY IS THE DENOMINATOR OF BAYES

Bayes' Theorem is just the multiplication rule applied twice and divided. Start from: $P(A_k \cap B) = P(B | A_k)P(A_k)$. Divide both sides by $P(B)$. The denominator $P(B) = \sum_i P(B | A_i)P(A_i)$ comes from total probability. The result is Bayes' Theorem.

THEOREM 3.2.1 – GENERAL FORM (ANY NUMBER OF CAUSES)

Bayes' Theorem

$$P(A_k | B) = \frac{P(B | A_k) P(A_k)}{\sum_{i=1}^n P(B | A_i) P(A_i)}$$

Prior: $P(A_k)$ — what you believed before observing B .

Likelihood: $P(B | A_k)$ — how probable the observation is if A_k is the cause.

Posterior: $P(A_k | B)$ — updated belief after observing B .

Normalising constant: $P(B)$ — ensures posteriors sum to 1.

SPECIAL CASE – TWO CAUSES

Bayes Two-Event Form

$$P(A | B) = \frac{P(B | A) P(A)}{P(B | A) P(A) + P(B | \bar{A}) P(\bar{A})}$$

STEP-BY-STEP METHOD

How to solve ANY Bayes question

Universal method — 5 steps:

1. **Identify the causes** $\{A_i\}$. These must be mutually exclusive and exhaustive. List all $P(A_i)$.
2. **Identify the observation** B . Write down all likelihoods $P(B | A_i)$.
3. **Compute the joint probabilities:** $P(A_i \cap B) = P(B | A_i) P(A_i)$ for each i . These are the "branch products" in a tree.
4. **Compute $P(B)$ by summing:** $P(B) = \sum_i P(A_i \cap B)$.
5. **Divide:** $P(A_k | B) = P(A_k \cap B) / P(B)$.

Tree diagram variant: Draw a tree with causes on level 1, outcomes on level 2. Multiply along branches for joint probabilities. Add all branches leading to B . Divide target branch by the sum. This is identical to steps 3–5 above.

Table method (for 3+ causes): Make a table with columns: Cause, Prior, Likelihood, Joint (= prior $\times$ likelihood), Posterior (= joint $\div$ sum of joints). Fill left to right. The sum of the Joint column is $P(B)$.

PATTERN RECOGNITION

How to identify a Bayes question

Universal trigger: You are asked to find the CAUSE from an OBSERVATION. If the question asks "given this result happened, which cause most likely produced it?" — that is Bayes.

PHRASING	BAYES?	CAUSE $\rightarrow$ OBSERVATION DIRECTION
"tested positive — probability they have disease?"	YES	+ result $\rightarrow$ disease
"item is defective — which machine made it?"	YES	defect $\rightarrow$ machine
"person coughs — probability they smoke?"	YES	cough $\rightarrow$ smoking
"passed the exam — probability she studied?"	YES	pass $\rightarrow$ studied
"email flagged as spam — is it actually spam?"	YES	flag $\rightarrow$ spam type
"a red card is drawn — probability it is a King?"	NO	No reversal — direct conditional
"machine has 3% defect rate — prob one item defective?"	NO	Cause $\rightarrow$ Effect direction, no reversal

THE TRAP

Base rate neglect — the most common Bayes error

Why it feels correct: "The test is 98% accurate → a positive test means 98% chance of disease." This ignores $P(\text{disease})$.

Why it is wrong: With a rare disease (3% prevalence), the large healthy population (97%) generates many more false positives than the tiny diseased population (3%) generates true positives. The base rate $P(\text{disease})$ cannot be ignored.

Concrete impact: Sensitivity = 98%, specificity = 93% (false positive rate 7%), prevalence 3%. Posterior =
$$\frac{0.98 \times 0.03}{0.98 \times 0.03 + 0.07 \times 0.97} = \frac{0.0294}{0.0973} \approx 30.2\%. \text{ Not } 98\%.$$

Second trap: Forgetting to compute $P(B)$. The denominator must include ALL paths to observation B . Missing even one partition makes the posterior wrong.

Third trap: Using causes that are not exhaustive. You MUST have $\sum P(A_i) = 1$. If three machines produce 40%, 40%, 10% of output, you're missing 10% of the partition — the computation is invalid.

WORKED EXAMPLES

Four complete Bayes solutions

Example 1 — Two machines (classic two-cause form):

Machine A: 40% of output, 10% defective. Machine B: 60%, 4% defective. Given defective item, find $P(M_A | D)$

- 1 Causes: $P(M_A) = 0.4$, $P(M_B) = 0.6$. Likelihoods: $P(D | M_A) = 0.10$, $P(D | M_B) = 0.04$.
- 2 Joint probabilities: $P(M_A \cap D) = 0.10 \times 0.4 = 0.040$; $P(M_B \cap D) = 0.04 \times 0.6 = 0.024$.
- 3 $P(D) = 0.040 + 0.024 = 0.064$.
- 4 $P(M_A | D) = 0.040/0.064 = \mathbf{0.625}$.

INTERPRETATION

Machine A makes only 40% of output but produces 62.5% of defectives because its defect rate is 2.5× higher. High likelihood can outweigh a low prior.

Example 2 — Three-cause (table method):

Suppliers: X (50%, 2% defect), Y (30%, 4%), Z (20%, 7%). Given defective, find $P(Z | D)$.

CAUSE	PRIOR $P(A_I)$	LIKELIHOOD $P(D A_I)$	JOINT $P(D A_I)P(A_I)$	POSTERIOR $P(A_I D)$
X	0.50	0.02	0.010	$0.010/0.036 = 0.278$
Y	0.30	0.04	0.012	$0.012/0.036 = 0.333$
Z	0.20	0.07	0.014	$0.014/0.036 = \mathbf{0.389}$
Total	1.00	—	0.036	1.000

Example 3 — Medical test (full four-part solution):

Disease in 3% of population. Test: sensitivity 98% (miss rate 2%), specificity 93% (false positive 7%).

- 1 $P(C) = 0.03$, $P(\bar{C}) = 0.97$. $P(+ | C) = 0.98$, $P(- | C) = 0.02$. $P(+ | \bar{C}) = 0.07$, $P(- | \bar{C}) = 0.93$.
- 2 $P(+) = 0.98 \times 0.03 + 0.07 \times 0.97 = 0.0294 + 0.0679 = 0.0973$.
- 3 $P(C | +) = 0.0294/0.0973 \approx \mathbf{0.302}$. Only ~30% of positive results are truly diseased!
- 4 $P(-) = 1 - 0.0973 = 0.9027$. $P(C | -) = 0.02 \times 0.03/0.9027 = 0.0006/0.9027 \approx \mathbf{0.000665}$.
Negative test almost certainly means healthy.

- 5 $P(\text{error}) = P(+ | \bar{C})P(\bar{C}) + P(- | C)P(C) = 0.0679 + 0.0006 \approx \mathbf{0.0685}$. About 6.85% of tests are incorrect.

Example 4 — Studied/Passed (most exam-style form):

$P(\text{pass} | \text{studied}) = 0.8$, $P(\text{pass} | \text{not studied}) = 0.3$, $P(\text{studied}) = 0.6$. Given passed, find $P(\text{studied} | \text{passed})$.

1 $P(\text{pass}) = 0.8(0.6) + 0.3(0.4) = 0.48 + 0.12 = 0.60$.

2 $P(\text{studied} | \text{passed}) = \frac{0.8 \times 0.6}{0.60} = \frac{0.48}{0.60} = \mathbf{0.80}$.

NOTE

It happens that the posterior equals the prior sensitivity. This is a numerical coincidence — verify it rather than assuming it.



A 98%-accurate test can mostly find healthy people

With 3% disease prevalence, even a 98%/93% test has a positive predictive value of only ~30%. About 70 out of every 100 people who test positive are actually healthy — false positives. The test is overwhelmingly more likely to flag a healthy person (97% of the population $\times$ 7%) than to correctly catch a diseased person (3% $\times$ 98%).

The lesson: For rare diseases, a single positive test is statistically unreliable. Confirmation testing is required. This is why doctors don't diagnose on a single test for rare conditions.

The reverse: A negative test result is extremely reliable ($P(C | -) \approx 0.07\%$). Negative tests rule things out far more cleanly than positive tests rule things in, when the condition is rare.

HOOK

Turning outcomes into numbers

A random variable is a translator. It takes the messy language of a random experiment ("HHT", "rolled a 3", "sour and fresh") and converts it into a real number so we can do algebra. Without this translation, we cannot compute expected values, variances, or use any formula from calculus or algebra.

MECHANICS — PMF, CDF, PDF, $E(X)$, $V(X)$

DEFINITION 4.1.1

Random Variable

A **random variable** $X : S \rightarrow \mathbb{R}$ is a function from the sample space to the real line. Use uppercase for the variable, lowercase for specific values: $P(X = x)$.

Discrete: takes countable (isolated) values — usually integers. Number of customers, defectives, heads.

Continuous: takes any value in an interval — time, distance, proportion, temperature.

4.3 Probability Mass Function (PMF) — Discrete

DEFINITION 4.3.1

PMF Requirements

A function $p(x) = P(X = x)$ is a valid PMF iff:

1. $p(x) \geq 0$ for all x in the support; $p(x) = 0$ outside the support.
2. $0 \leq p(x) \leq 1$ for all x .
3. $\sum_{\text{all } x} p(x) = 1$.

Strict vs non-strict inequalities (discrete only): $P(X \geq 3) \neq P(X > 3)$. They differ by exactly $p(3)$. For continuous RVs, $P(X = c) = 0$ so strict/non-strict distinctions vanish.

DISCRETE INTERVAL

Probability on a Range

$$P(a \leq X \leq b) = \sum_{x=a}^b p(x)$$

Watch boundaries! $P(X > 3) = P(X \geq 4) = \sum_{x=4}^{\infty} p(x)$
for integer-valued X .

COMPLEMENT

At-Least Problems

$$P(X \geq k) = 1 - P(X \leq k-1) = 1 - \sum_{x=0}^{k-1} p(x)$$

Use when "at least k " requires summing many terms — complement is often shorter.

4.3.1 Cumulative Distribution Function (CDF)

DEFINITION 4.3.2

CDF — Properties and Reading

$F(a) = P(X \leq a)$. For discrete X : $F(a) = \sum_{x \leq a} p(x)$. A **step function** — flat between mass points, jumping by $p(x_i)$ at each mass point.

Four key properties:

1. Non-decreasing: $\backslash(a$
2. $\lim_{x \rightarrow \infty} F(x) = 1$.
3. $\lim_{x \rightarrow -\infty} F(x) = 0$.
4. Right-continuous: $\lim_{a_n \downarrow a} F(a_n) = F(a)$.

READING THE CDF

Recovering Probabilities

$$P(X = a) = F(a) - F(a^-) \text{ (the jump size at } a)$$

$$P(a < X \leq b) = F(b) - F(a)$$

$$P(X < a) = F(a^-) \quad P(X > a) = 1 - F(a)$$

$F(a^-)$ = value just below a , i.e. limit from the left.

CDF EXAMPLE

Step Function

If

$$p(1) = 1/4, p(2) = 1/2, p(3) = 1/8, p(4) = 1/8$$

:

$$F(x) = \begin{cases} 0 & x < 1 \\ 1/4 & 1 \leq x < 2 \\ 3/4 & 2 \leq x < 3 \\ 7/8 & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

CRITICAL — CDF AT JUMP POINTS

For discrete X : $P(X < 3) = F(2) \neq F(3) = P(X \leq 3)$. These differ by $p(3)$. When reading a CDF: "strictly less than 3" means use the value just before the jump at 3.

4.3.2 Probability Density Function (PDF) — Continuous

DEFINITION 4.3.3

PDF — What It Is and Isn't

A function $f(x)$ is a valid PDF iff: (1) $f(x) \geq 0$ for all x , (2) $\int_{-\infty}^{\infty} f(x) dx = 1$.

$P(a \leq X \leq b) = \int_a^b f(x) dx$. For continuous X : $P(X = c) = 0$ for any fixed c .

CRITICAL: $f(x)$ is NOT a probability. PDF values can exceed 1. Only the *integral* (area) must equal 1. E.g. $f(x) = 2x$ on $[0, 1]$ gives $f(0.9) = 1.8 > 1$ — perfectly valid.

CDF from PDF: $F(a) = \int_{-\infty}^a f(x) dx$. **PDF from CDF:** $f(x) = F'(x)$. For continuous RVs:

$P(X < a) = P(X \leq a) = F(a)$ — strict/non-strict distinction vanishes.

FINDING K — NORMALISATION

Normalisation Condition

Set $\int_{-\infty}^{\infty} f(x) dx = 1$ and solve for unknown constant k .

Integrate over the full support (where $f(x) \neq 0$), not all of $\mathbb{R}$ unless support is all reals.

COMPUTING PROBABILITIES

Interval Probabilities (Continuous)

$$P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

For continuous RVs, $P(a < X < b) = P(a \leq X \leq b)$ — endpoints have zero probability.

4.4 Expectation & Variance — Full Treatment

§4.4.1 — THE PROBABILITY-WEIGHTED AVERAGE

Expectation (Mean)

Discrete: $E(X) = \sum_{x: p(x) > 0} x p(x)$

Continuous: $E(X) = \int_{-\infty}^{\infty} x f(x) dx$

The expected value is the probability-weighted average. It need not be a value X can actually take (e.g. $E(\text{die roll}) = 3.5$).

LAW OF THE UNCONSCIOUS STATISTICIAN

Expectation of a Function $E[g(X)]$

Discrete: $E[g(X)] = \sum_x g(x) p(x)$

Continuous: $E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$

Key uses: $g(X) = X^2$ to get $E(X^2)$ for variance; $g(X) = aX + b$ for linear transforms.

PROPOSITION 4.4.1

Linearity of Expectation

$$E(aX + b) = aE(X) + b$$

$$E(c) = c \text{ for any constant } c$$

$$E(X - \mu) = 0 \text{ where } \mu = E(X)$$

USEFUL IDENTITY

$E(X^2)$ from Variance

$$E(X^2) = V(X) + [E(X)]^2$$

Rearrangement of the computational formula for variance.

§4.4.2 – SPREAD AROUND THE MEAN

Variance & Standard Deviation

$$V(X) = E[(X - \mu)^2] = E(X^2) - \mu^2 \text{ where } \mu = E(X)$$

Discrete: $V(X) = \sum_x (x - \mu)^2 p(x)$

Continuous: $V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$

$SD(X) = \sqrt{V(X)}$. $V(aX + b) = a^2 V(X)$. $V(c) = 0$. *The computational formula $E(X^2) - \mu^2$ is almost always faster than the definition.*

QUANTITY	DISCRETE FORMULA	CONTINUOUS FORMULA
$E(X)$	$\sum x p(x)$	$\int x f(x) dx$
$E[g(X)]$	$\sum g(x) p(x)$	$\int g(x) f(x) dx$
$E(X^2)$	$\sum x^2 p(x)$	$\int x^2 f(x) dx$
$V(X)$	$E(X^2) - [E(X)]^2$	
$E(aX + b)$	$aE(X) + b$	
$V(aX + b)$	$a^2 V(X)$	

QUANTITY	DISCRETE FORMULA	CONTINUOUS FORMULA
$V(c)$	0	

ⓘ CONTRAST PMF vs PDF — similar name, different nature

- **PMF $p(x)$:** IS a probability. Must be in $[0, 1]$. Sum of all values = 1. Individual point has non-zero probability.
- **PDF $f(x)$:** is NOT a probability. Can exceed 1. Integral over all reals = 1. Individual point has zero probability.
- **Strict inequalities:** For discrete — $P(X \geq 3) \neq P(X > 3)$ (differ by $p(3)$). For continuous — $P(X \geq 3) = P(X > 3)$ (since $P(X = 3) = 0$).

ⓘ THE TRAP Four common random variable mistakes

1. **Using definition formula instead of computational formula for $V(X)$:** Computing $\sum (x - \mu)^2 p(x)$ term by term when $E(X^2) - \mu^2$ is much faster. Always use the computational formula first.
2. **Not finding k before computing $E(X)$:** If the PDF has an unknown constant, you MUST find it via the normalisation condition before any other calculation. An unnormalised PDF gives wrong expectations.
3. **Confusing discrete boundary cases:** $P(X < 2) \neq P(X \leq 2)$ for discrete X . The difference is $p(2)$. Always write out the exact summation explicitly.
4. **Computing $E[g(X)]$ as $g[E(X)]$:** $E(X^2) \neq [E(X)]^2$ in general (they differ by the variance). LOTUS requires summing $g(x) \cdot p(x)$, not plugging $E(X)$ into g .

ⓘ WORKED EXAMPLES

Example A — Finding k and computing E(X), V(X) from a PDF:

$f(x) = \frac{3}{4}(kx - x^2)$ on $[0, 2]$, zero otherwise.

- 1 Normalise: $\int_0^2 \frac{3}{4}(kx - x^2)dx = \frac{3}{4} \left[k\frac{x^2}{2} - \frac{x^3}{3} \right]_0^2 = \frac{3}{4}(2k - \frac{8}{3}) = 1$. Solving: $2k = \frac{4}{3} + \frac{8}{3} = 4 \Rightarrow k = 2$.
- 2 So $f(x) = \frac{3}{4}(2x - x^2) = \frac{3}{2}x - \frac{3}{4}x^2$ on $[0, 2]$.
- 3 $E(X) = \int_0^2 x(\frac{3}{2}x - \frac{3}{4}x^2)dx = \int_0^2 (\frac{3}{2}x^2 - \frac{3}{4}x^3)dx = \left[\frac{1}{2}x^3 - \frac{3}{16}x^4 \right]_0^2 = 4 - 3 = 1$.
- 4 $E(X^2) = \int_0^2 x^2(\frac{3}{2}x - \frac{3}{4}x^2)dx = \left[\frac{3}{8}x^4 - \frac{3}{20}x^5 \right]_0^2 = 6 - 4.8 = 1.2$.
- 5 $V(X) = E(X^2) - [E(X)]^2 = 1.2 - 1 = 0.2$.

Example B — Finding PMF from a CDF:

$F(x) = 0$ ($x < 0$); $\frac{1}{2}$ ($0 \leq x < 1$); $\frac{3}{5}$ ($1 \leq x < 2$); $\frac{4}{5}$ ($2 \leq x < 3$); $\frac{9}{10}$ ($3 \leq x < 4$); 1 ($x \geq 4$).

- 1 $p(0) = F(0) - F(0^-) = \frac{1}{2} - 0 = \frac{1}{2}$
- 2 $p(1) = F(1) - F(1^-) = \frac{3}{5} - \frac{1}{2} = \frac{1}{10}$
- 3 $p(2) = \frac{4}{5} - \frac{3}{5} = \frac{1}{5}$; $p(3) = \frac{9}{10} - \frac{4}{5} = \frac{1}{10}$; $p(4) = 1 - \frac{9}{10} = \frac{1}{10}$
- 4 Check: $\frac{1}{2} + \frac{1}{10} + \frac{1}{5} + \frac{1}{10} + \frac{1}{10} = \frac{10}{10} = 1$. ✓

Example C — Softball player: finding missing probabilities then E(X), V(X):

$P(X = 1) = 0.3$, $P(X = 2) = 0.2$, $P(X = 0) = 3P(X = 3)$.

- 1 Let $P(X = 3) = q$. Then $P(X = 0) = 3q$. Total: $3q + 0.3 + 0.2 + q = 1 \Rightarrow 4q = 0.5 \Rightarrow q = 0.125$.
- 2 $P(X = 0) = 0.375$, $P(X = 3) = 0.125$.
- 3 $E(X) = 0(0.375) + 1(0.3) + 2(0.2) + 3(0.125) = 0 + 0.3 + 0.4 + 0.375 = 1.075$.
- 4 $E(X^2) = 0 + 1(0.3) + 4(0.2) + 9(0.125) = 0.3 + 0.8 + 1.125 = 2.225$.
- 5 $V(X) = 2.225 - 1.075^2 = 2.225 - 1.1556 = 1.069$.

Consider: $W \equiv 0$ always; $Y = \pm 1$ each with prob $\frac{1}{2}$; $Z = \pm 100$ each with prob $\frac{1}{2}$. All three have $E = 0$. But $V(W) = 0$, $V(Y) = 1$, $V(Z) = 10,000$. The mean alone tells you nothing about spread — a savings account and a stock portfolio can have the same expected return but wildly different risk (variance).

CHAPTER 05 Discrete Probability Distributions

HOOK Named distributions are pre-solved models

Once you recognise that your problem fits a known pattern, the formula gives the answer immediately. The skill is recognising which distribution applies — and verifying every condition before using it.

5.1 Binomial Distribution $X \sim \text{Bin}(n, p)$

§5.1 – FOUR CONDITIONS (ALL MUST HOLD)

Binomial Setup

1. Fixed number of trials n .
2. Each trial is *independent*.
3. Each trial has the same probability of success p .
4. Each trial has exactly two outcomes (success / failure).

If ANY condition fails: (a) sampling without replacement from small population $\rightarrow$ Hypergeometric; (b) n is not fixed (counting until success) $\rightarrow$ Geometric; (c) p varies $\rightarrow$ not Binomial.

DEFINITION 5.1.2

Binomial PMF, Mean, Variance

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$E(X) = np \quad V(X) = np(1-p)$$

$\binom{n}{x}$ counts sequences with exactly x successes; $p^x(1-p)^{n-x}$ is the probability of each sequence. If $p = 0$: $P(X = 0) = 1$. If $p = 1$: $P(X = n) = 1$. Both give $V(X) = 0$.

COMPLEMENTARY

At Least k

$$P(X \geq k) = 1 - P(X \leq k-1) = 1 - \sum_{x=0}^{k-1} \binom{n}{x} p^x (1-p)^{n-x}$$

Always faster when "at least k " requires many terms.

SYMMETRY

Complementing Success/Failure

If $X \sim \text{Bin}(n, p)$ counts successes, then $n - X \sim \text{Bin}(n, 1-p)$ counts failures.

When $p > 0.5$: easier to count failures ($q = 1-p$) using the flipped distribution.

BINOMIAL VS HYPERGEOMETRIC – SAMPLING WITH VS WITHOUT REPLACEMENT

Binomial: sampling WITH replacement OR from a large population ($n < 5\%$ of population). p stays constant.

Hypergeometric: sampling WITHOUT replacement from a small finite population. p changes on each draw. If $n \geq 5\%$ of population, use Hypergeometric exactly.

5.2 Poisson Distribution $X \sim \text{Poi}(\lambda)$

§5.2 – THREE POISSON CONDITIONS

Poisson Process

1. Events occur randomly and independently of one another.
2. Average rate λ per unit interval is constant over time/space.
3. Two events cannot occur at exactly the same instant (no simultaneous events).

Use: arrivals, failures, accidents, calls per unit time. **Key diagnostic:** if $E(X) \approx V(X)$ in sample data, Poisson is plausible.

DEFINITION 5.2.1

Poisson PMF, Mean, Variance

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$E(X) = \lambda \quad V(X) = \lambda$$

Mean = Variance = λ . Extremely distinctive — no other standard distribution has this property.

SCALING

Changing the Interval

Rate λ per unit, interval of length t : $X \sim \text{Poi}(\lambda t)$

e.g. 12 failures per week = 2.4 per day. Over 3 days:

$$X \sim \text{Poi}(7.2).$$

ADDITIVE PROPERTY

Sum of Independent Poissons

If $X \sim \text{Poi}(\lambda_1) \perp Y \sim \text{Poi}(\lambda_2)$:

$$X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$$

COMMON FORM

At Least One

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\lambda}$$

APPROXIMATION RULE

Poisson $\approx$ Binomial

Use when: $n \geq 20, p \leq 0.05$ (or $n \geq 100, np \leq 10$). Set $\lambda = np$.

Exact limit: $\text{Bin}(n, p) \rightarrow \text{Poi}(\lambda)$ as

$$n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda.$$

5.3 Geometric & Negative Binomial

§5.3 - GEOMETRIC DISTRIBUTION $X \sim G(p)$

X = failures before **FIRST** success

$$p(x) = p(1-p)^x, \quad x = 0, 1, 2, \dots$$

$$E(X) = \frac{1-p}{p} \quad V(X) = \frac{1-p}{p^2}$$

Alternative parameterisation (trials until first success): $T = X + 1 \sim \text{Geometric}$ with

$$P(T = t) = p(1-p)^{t-1}, \quad t = 1, 2, \dots \text{ Make sure you know which version is being used.}$$

Memoryless property (only discrete distribution with this): $P(X > m + n \mid X > m) = P(X > n)$. Past failures are irrelevant — your next trial has the same probability as your first.

Tail probability: $P(X \geq k) = (1-p)^k$. Useful shortcut: $P(\text{no success in first } k \text{ trials}) = (1-p)^k$.

§5.3 – NEGATIVE BINOMIAL $X \sim \text{NEGBIN}(R, P)$

X = failures before the r -th success

$$p(x) = \binom{x+r-1}{x} p^r (1-p)^x, \quad x = 0, 1, 2, \dots$$

$$E(X) = \frac{r(1-p)}{p} \quad V(X) = \frac{r(1-p)}{p^2}$$

Geometric is special case $r = 1$. For large r , approximate using the Normal distribution: standardise $(X - \mu)/\sigma$.

5.4 Hypergeometric Distribution

§5.4 – HYPERGEOMETRIC

X = number of "successes" in sample drawn **WITHOUT** replacement

Population: N items, K "successes", $N - K$ "failures". Draw n without replacement.

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, \dots, \min(n, K)$$

$$E(X) = \frac{nK}{N}$$

When $n < 5\%N$: Binomial is acceptable approximation. When sampling from small populations: must use Hypergeometric.

ⓘ CONTRAST

Which distribution? — Decision guide

SCENARIO	DISTRIBUTION	PARAMETER(S)
Single yes/no trial	Bernoulli	p
n independent yes/no trials, count successes	Binomial	n, p
Count events in fixed interval, random independent arrivals	Poisson	λ
Count FAILURES before first success	Geometric	p
Count FAILURES before r -th success	Negative Binomial	r, p
Sample n WITHOUT replacement from finite population	Hypergeometric	N, K, n
Binomial-like but $n \geq 20, p \leq 0.05$	Poisson approx.	$\lambda = np$

Binomial checklist (all four must hold): (1) Fixed n . (2) Independent trials. (3) Constant p . (4) Two outcomes. If n is what you're counting $\rightarrow$ Geometric/NegBin. If sampling without replacement $\rightarrow$ Hypergeometric.

THE TRAP

Three distribution misapplication errors

- Applying Binomial when sampling without replacement:** Each draw changes p . Use Hypergeometric unless $n < 5\%N$.
- Wrong Poisson rate:** If the rate is given per week but the question asks about a 3-day period, $\lambda t = (\lambda_{\text{week}}/7) \times 3$. Always match units.
- Geometric parameterisation confusion:** X = failures before success ($x = 0, 1, 2, \dots$) vs T = trial number of first success ($t = 1, 2, 3, \dots$). These give different PMFs and different expectations. Read the question carefully.

WORKED EXAMPLES

Example A — True/false test (Binomial): 5 questions, guess randomly. Pass if ≥ 4 correct. $X \sim \text{Bin}(5, 0.5)$.

1 $P(X = 4) = \binom{5}{4} (0.5)^5 = 5/32$. $P(X = 5) = \binom{5}{5} (0.5)^5 = 1/32$.

2 $P(X \geq 4) = 6/32 = \mathbf{0.1875}$.

Example B — Ice cream vendor (Poisson, stock level): Sales average 10/hour. $X \sim \text{Poi}(10)$. Find smallest k with $P(X \leq k) \geq 0.95$.

1 Compute $P(X \leq k)$ incrementally using $p(x) = e^{-10} 10^x / x!$.

2 $P(X \leq 14) \approx 0.9165 < 0.95$. $P(X \leq 15) \approx 0.9513 \geq 0.95$.

3 Stock needed: **15 ice creams**.

Example C — Sour milk (Hypergeometric): 10 cartons, 3 sour. Select 4. $P(X \geq 2)$.

1 $\binom{10}{4} = 210$.

2 $P(X = 2) = \binom{3}{2} \binom{7}{2} / 210 = 3 \times 21 / 210 = 63 / 210$.

3 $P(X = 3) = \binom{3}{3} \binom{7}{1} / 210 = 1 \times 7 / 210 = 7 / 210$.

4 $P(X \geq 2) = (63 + 7) / 210 = 70 / 210 = \mathbf{1/3} \approx 0.333$.

Example D — Die until 6 (Geometric): $p = 1/6$. Probability no 6 in first 10 tosses.

1 $P(X \geq 10) = (1 - 1/6)^{10} = (5/6)^{10} \approx \mathbf{0.1615}$.

Example E — Negative Binomial (large r , Normal approximation): 50 signatures needed, each delegate signs with prob 0.15. $P(\geq 300 \text{ delegates approached}) = P(X \geq 250 \text{ failures before 50th success})$.

1 $X \sim \text{NegBin}(50, 0.15)$. $E(X) = 50(0.85)/0.15 \approx 283.3$. $V(X) = 50(0.85)/0.15^2 \approx 1888.9$. $\sigma \approx 43.5$.

2 Normal approximation: $z = (250 - 283.3)/43.5 \approx -0.765$.

3 $P(X \geq 250) \approx \Phi(0.765) \approx \mathbf{0.778}$.



The memoryless property: past failures are irrelevant

$P(X > m + n \mid X > m) = P(X > n)$. If you've tossed a die 10 times without getting a 6, the 11th toss has exactly the same probability of being a 6 as the first one. The die has no memory. The Geometric distribution is the *only* discrete distribution with this property.

This is counterintuitive to people who believe in "hot streaks" or "due for a win." The gambler's fallacy is the false belief that past outcomes affect future probabilities in independent trials.



Complete Exam Traps & Pitfalls

Every known error type — know these before you sit the exam.

SET THEORY • 1



Confusing $\subset$ with $\in$

✗ $D \in C$ when $D = \{2, 4\}$, $C = \{1, 2, 3, 4\}$

✓ $D \subset C$ — a set is a subset, not an element

$\in$ is for single objects. $\subset$ is for sets-within-sets.

SET THEORY • 2



Discrete vs continuous set equality

✗ $\{1, 2, 3, 4\} = \{x \mid 1 \leq x \leq 4\}$

✓ $\{1, 2, 3, 4\}$ (4 integers) $\neq$ $[1, 4]$ (uncountably many reals)

Always check: listing integers vs continuous interval.

SET THEORY • 3



De Morgan direction

✗ $\overline{A \cup B} = \bar{A} \cup \bar{B}$

✓ $\overline{A \cup B} = \bar{A} \cap \bar{B}$ — operator flips

"Break the bar, change the operator." $\cup \leftrightarrow \cap$.

PROBABILITY • 4



Forgetting the intersection in the addition rule

✗ $P(A \cup B) = P(A) + P(B)$

✓ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Only omit subtraction when mutually exclusive ($P(A \cap B) = 0$).



Impossible probability combinations

✗ Accepting $P(A \cap B) > P(A)$ or
 $P(A \cup B) < \max(P(A), P(B))$

✓ Always check: $P(A \cap B) \leq \min(P(A), P(B))$
 and $P(A \cup B) \geq \max(P(A), P(B))$

These are necessary conditions from monotonicity. Check BEFORE computing.



Reversing $P(A | B)$ and $P(B | A)$

✗
 $P(\text{Machine A} | \text{defect}) = P(\text{defect} | \text{Machine A})$

✓ These are generally unequal. Use Bayes to
 convert one to the other.

The Prosecutor's Fallacy. Identify: what is given? What is sought?



Independence $\neq$ Mutual Exclusivity

✗ $P(A \cap B) = 0 \Rightarrow A$ and B are independent

✓ Mutual exclusivity ($P(A \cap B) = 0$) implies
 DEPENDENCE for non-trivial events

Independence requires $P(A \cap B) = P(A)P(B)$, not zero.



Ignoring base rates

✗ 98% accurate test $\rightarrow$ positive means 98% chance
 of disease

✓ With 3% prevalence: $P(\text{disease} | +) \approx 30\%$.
 Always include $P(A_k)$.

*Low prior probability can dominate even high likelihoods.
 Never skip the denominator.*



Non-exhaustive partition

✗ Using
 $P(B) = P(B | A_1)P(A_1) + P(B | A_2)P(A_2)$
 when $P(A_1) + P(A_2) < 1$

✓ All causes must sum to 1: $\sum_i P(A_i) = 1$

Missing a partition makes $P(B)$ too small, inflating all posteriors.



Order matters vs not

✗ Using $\binom{n}{r}$ when order is significant

✓ Use ${}_nP_r$ when order matters. Ask: "Does
 swapping two items give a different outcome?"

*PIN codes, race placements $\rightarrow$ permutation. Committees,
 hands $\rightarrow$ combination.*



Forgetting position-by-position restrictions

✗ Assuming each non-first digit has 9 choices after fixing a digit

✓ Account for ALL restrictions applied so far: if digit 1=7 and digit 2≠0, choices for digit 2 = $10 - 2 = 8$

Work position by position. Track which digits have been excluded at each step.



PDF value > 1 seems invalid

✗ $f(0.9) = 1.8$ means the PDF is wrong

✓ PDF values are NOT probabilities — they can exceed 1. The INTEGRAL must equal 1.

$f(x) = 2x$ on $[0, 1]$: $f(0.9) = 1.8 > 1$, but $\int_0^1 2x \, dx = 1$. ✓
Perfectly valid.



Strict vs non-strict inequalities in discrete RVs

✗ $P(X \geq 3) = P(X > 3)$

✓ For discrete: differ by $p(3)$. For continuous: equal (since $P(X = c) = 0$).

*Always write out the exact summation for discrete RVs.
Never assume the boundary is negligible.*



$$E[g(X)] \neq g[E(X)]$$

✗ $E(X^2) = [E(X)]^2$

✓ $E(X^2) = V(X) + [E(X)]^2$. They differ by the variance.

ALWAYS use $E(X^2) = \sum x^2 p(x)$ (or integral). Never plug $E(X)$ into g .



Binomial without replacement from small population

✗ Using Binomial when sampling from 10 items without replacement

✓ Use Hypergeometric. Binomial requires independence; sampling changes p .

Rule: $n \geq 5\%N \rightarrow$ must use Hypergeometric. $n < 5\%N \rightarrow$ Binomial approximation acceptable.



Wrong Poisson time interval

✗ Rate is 12/week, question asks about 3 days: use $\lambda = 12$

✓ Convert: $\lambda = 12/7$ per day, so over 3 days:
 $\lambda t = (12/7) \times 3 \approx 5.14$

Always match rate and interval units before computing.



Problem Sheets 1–9 — All Questions

Set Theory · Probability · Conditional · Bayes · Random Variables · Distributions

PROBLEM SHEET 1 — SET THEORY

Q1

EASY

Let $U = \{1, 2, \dots, 10\}$, $A = \{1, \dots, 8\}$, $B = \{2, 3, 4\}$, $C = \{1, 3, 5, 7, 9\}$.

1. Find $\bar{A}$, $B \cup \bar{C}$, and $A \cap (\bar{B} \cup C)$.
2. Is it true that $\bar{B} \subset A \cap C$?

Q2

MEDIUM

Let $A = \{x \mid 0$

1. $C = E$
2. $D \subset C$
3. $D \in E$
4. $E \not\subset A$

Q3

HARD

Prove algebraically that $(A - B) - C = A - (B \cup C)$ using set identities.

Q4

MEDIUM

Let S = all companies on the Botswana Stock Exchange. A =mining sector, B =turnover >P10m, C =financial year ending June, D =share price higher than 6 months ago. Describe in words the set $B \cap (C \cup A)$.

PROBLEM SHEET 2 – PROBABILITY FOUNDATIONS

Q5

EASY

$P(A) = 0.3$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Find the probability of:

1. Either A or B occurs.
2. A does not occur.
3. Neither A nor B occurs.
4. Exactly one of A or B occurs.

Q6

MEDIUM

Which combinations are *impossible*?

1. $P(A) = 0.7$, $P(B) = 0.9$, $P(A \cap B) = 0.4$
2. $P(A) = 0.2$, $P(B) = 0.5$, $P(A \cap B) = 0.25$
3. $P(A) = 0.8$, $P(B) = 0.4$, $P(A \cup B) = 0.7$

Q7

HARD

Show that $P(A \cup \bar{B}) = 1 - [P(B) - P(A \cap B)]$.

PROBLEM SHEET 3 – COMBINATORIAL PROBABILITY

Q8

MEDIUM

An 8-digit cellphone number starts with 7 and has second digit $\neq 0$. If generated with no repeated digits, what is the probability of this being satisfied if we just require the two constraints (start 7, second $\neq 0$)?

Q9

HARD

A pack of 52 cards dealt into four bridge hands of 13. Find probability a given hand contains:

1. 3 spades, 4 diamonds, 1 heart, 5 clubs.
2. Exactly 3 aces and exactly 4 picture cards (J, Q, K).

Q10

MEDIUM

75 students divided into tutorial groups of 25, 30, 20.

1. In how many ways can this be done?
2. Two brothers are in the class. Probability they end up in the same group?

PROBLEM SHEET 4 – CONDITIONAL PROBABILITY & BAYES

Q11

EASY

$P(A) = 0.6$, $P(B) = 0.2$, $P(A | B) = 0.5$. Find:

1. $P(B | A)$
2. $P(A \cup B)$
3. $P(\bar{A} \cap B)$
4. $P(B | \bar{A})$

Q12

MEDIUM

Machine A: 40% of output, 10% defective. Machine B: 60%, 4% defective.

1. What proportion of all output is defective?
2. Given defective, probability it came from Machine A?

Q13

HARD

X-ray test for disease present in 3% of population. False negative rate 2%, false positive rate 7%.

1. Proportion of diseased people correctly diagnosed?
2. Of those positive, proportion truly diseased?
3. Of those negative, proportion truly diseased?
4. Proportion of tests giving incorrect diagnosis?

Q14

MEDIUM

$P(A) = 0.6$, $P(B | A) = 0.3$, $P(A \cup B) = 0.72$. Are A and B independent, mutually exclusive, or neither?

Q15

HARD

Student passes with probability 0.8 if studied, 0.3 if not. 60% of class studied. Given a student passed, probability she studied?

PROBLEM SHEET 5 – DEFINING RANDOM VARIABLES

Q16

EASY

State whether each is a RV; if so, discrete or continuous, and write the range:

1. Time to answer this question.
2. Words in a randomly chosen library book.
3. Maximum temperature at Sir Seretse Khama Airport.
4. Application letters before landing a job.
5. Proportion of students passing STA122.

Q17

MEDIUM

A die rolled twice. Find values and probabilities for:

1. Maximum of both rolls.
2. Sum of both rolls.
3. First minus second roll.

Q18

MEDIUM

Urn: 6 white, 2 black, 2 red. Draw 2. Win P20 per black, lose P10 per white (red: no effect). X = total winnings. Find all values and probabilities.

Q19

HARD

Pool company completes 2 or 3 contracts/month equally likely. Contracts received:

$P(1) = 0.1$, $P(2) = 0.2$, $P(3) = 0.5$, $P(4) = 0.2$. 2 carried forward from last month. Let X = contracts carried forward to next month. Find distribution of X .

PROBLEM SHEET 6 – PMF AND CDF

Q20

MEDIUM

Which satisfy the PMF conditions?

1. $p(x) = 0.25(0.75)^x$, $x = 0, 1, 2, \dots$
2. $p(x) = \frac{1}{5}(2x - 3)$, $x = 0, 1, 2, 3, 4, 5$
3. $p(x) = \binom{5}{x} \binom{4}{x-2} / 36$, $x = 0, 1, 2$

Q21

MEDIUM

$p(x) = k/x!$ for $x = 0, 1, 2, 3, 4$, zero otherwise. (i) Find k . (ii) Find $P(X < 2)$.

Q22

MEDIUM

CDF: $F(x) = 0$ ($x < 0$), $\frac{1}{2}$ ($0 \leq x < 1$), $\frac{3}{5}$ ($1 \leq x < 2$), $\frac{4}{5}$ ($2 \leq x < 3$), $\frac{9}{10}$ ($3 \leq x < 4$), 1 ($x \geq 4$).

Find the PMF.

PROBLEM SHEET 7 – EXPECTATION AND VARIANCE

Q23

MEDIUM

$P(X = 1) = 0.3, P(X = 2) = 0.2, P(X = 0) = 3P(X = 3)$. Find $E(X)$ and $V(X)$.

Q24

HARD

Using the pool company distribution from Q19, find $E(X)$ and $V(X)$.

Q25

HARD

$f(x) = \frac{3}{4}(kx - x^2)$ for $0 \leq x \leq 2$, zero elsewhere. Find $k, E(X), V(X)$.

PROBLEM SHEET 8 – BINOMIAL AND POISSON

Q26

EASY

True-false test, 5 questions. Pass with ≥ 4 correct. If guessing, probability of passing?

Q27

MEDIUM

Engines fail independently with probability 0.1. Successful flight if at most half fail. Probability for: (i) 2-engine, (ii) 4-engine plane.

Q28

MEDIUM

Right-turn lane: capacity 3 cars. 30% want to turn right. 15 cars approach. Probability lane is insufficient?

Q29

MEDIUM

Ice cream: Poisson, average 10/hour. (i) Probability at least one sale in first hour. (ii) Stock size for 95% certainty of not running out?

Q30

MEDIUM

Pump fails once per 500 hours on average. Probability exactly 3 failures in 2000 hours?

Q31

HARD

Quality control: test 10 items, reject if ≥ 1 defective. (i) Find rejection probs for $p = 1\%, 10\%, 30\%$.
(ii) Compare to: test 20, reject if ≥ 2 . (iii) Which procedure is better?

Q32

MEDIUM

Average demand: 8 motors/week; delivery 1 week. Poisson demand. How low should stock fall before ordering for 95% certainty of meeting demand?

Q33

MEDIUM

Fair die until 6 appears. (i) Probability no 6 in first 10 tosses. (ii) First 6 exactly on 6th toss.

Q34

HARD

Candidate needs 50 signatures. Each delegate signs with prob 0.15. Probability that ≥ 300 delegates must be approached?

Q35

MEDIUM

Tuckshop: 10 milk cartons, 3 sour. Select 4. Probability at least 2 sour? (Hypergeometric.)

Full Solutions — All Questions

Complete worked solutions Q1–Q35.

SOLUTIONS – SHEET 1

A1

1. $\bar{A} = \{9, 10\}$

2. $\bar{C} = \{2, 4, 6, 8, 10\}$. $B \cup \bar{C} = \{2, 3, 4, 6, 8, 10\}$

3. $\bar{B} = \{1, 5, 6, 7, 8, 9, 10\}$. $\bar{B} \cup C = \{1, 3, 5, 6, 7, 8, 9, 10\}$. $A \cap (\bar{B} \cup C) = \{1, 3, 5, 6, 7, 8\}$

4. $A \cap C = \{1, 3, 5, 7\}$. Since $6 \in \bar{B}$ but $6 \notin A \cap C$: **FALSE**

A2

1. $\{1, 2, 3, 4\}$ vs $[1, 4]$: **FALSE**

2. $\{2, 4\} \subset \{1, 2, 3, 4\}$: **TRUE**

3. $\{2, 4\}$ is a set, not a real number: **FALSE**

4. Every $x \in [1, 4]$ satisfies $\neg(0$ **FALSE**

A3

$$(A - B) - C = (A \cap \bar{B}) \cap \bar{C} = A \cap (\bar{B} \cap \bar{C}) = A \cap \overline{(B \cup C)} = A - (B \cup C). \quad \square$$

A4

Companies with turnover >P10m that are either in the mining sector OR have a financial year ending in June (or both).

A5

$$1. P(A \cup B) = 0.3 + 0.4 - 0.2 = \boxed{0.5}$$

$$2. P(\bar{A}) = 1 - 0.3 = \boxed{0.7}$$

$$3. P(\overline{A \cup B}) = 1 - 0.5 = \boxed{0.5}$$

$$4. P(\text{exactly one}) = P(A \cup B) - P(A \cap B) = 0.5 - 0.2 = \boxed{0.3}$$

A6

$$1. P(A \cap B) = 0.4 \leq \min(0.7, 0.9) = 0.7. \checkmark \quad \boxed{\text{Possible.}}$$

$$2. P(A \cap B) = 0.25 > P(A) = 0.2. \text{ Violates } A \cap B \subset A. \quad \boxed{\text{Impossible.}}$$

$$3. P(A \cup B) = 0.7 < \max(0.8, 0.4) = 0.8. \text{ Violates monotonicity.} \quad \boxed{\text{Impossible.}}$$

A7

By partition rule: $P(B \cap \bar{A}) = P(B) - P(A \cap B)$. By De Morgan:

$$P(A \cup \bar{B}) = 1 - P(\overline{A \cup \bar{B}}) = 1 - P(\bar{A} \cap B) = 1 - [P(B) - P(A \cap B)]. \quad \square$$

A8

Digit 1 fixed (7): 1 choice. Digit 2 ($\neq 0, \neq 7$): 8 choices. Digits 3–8 (no repeats): $8 \times 7 \times 6 \times 5 \times 4 \times 3$.

Favourable: $1 \times 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3$. Total (only constraint: start 7):

$$1 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3.$$

$$P = \frac{8}{9} \approx$$

$$\boxed{0.8889}$$

A9

$$1. \frac{\binom{13}{3} \binom{13}{4} \binom{13}{1} \binom{13}{5}}{\binom{52}{13}} \approx \mathbf{0.00539}$$

$$2. \frac{\binom{4}{3} \binom{12}{4} \binom{36}{6}}{\binom{52}{13}} \approx \mathbf{0.00412}$$

A10

$$1. \binom{75}{25,30,20} = 75! / (25! 30! 20!) \approx 5.99 \times 10^{35}$$

$$2. P = \frac{25 \cdot 24 + 30 \cdot 29 + 20 \cdot 19}{75 \cdot 74} = \frac{1850}{5550} = \frac{1}{3} \approx \mathbf{0.333}$$

SOLUTIONS – SHEET 4

A11

$$1. P(A \cap B) = 0.5 \times 0.2 = 0.10. P(B | A) = 0.10/0.60 = \frac{1}{6} \approx \mathbf{0.167}$$

$$2. P(A \cup B) = 0.6 + 0.2 - 0.1 = \mathbf{0.7}$$

$$3. P(\bar{A} \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = \mathbf{0.1}$$

$$4. P(B | \bar{A}) = 0.1/0.4 = \mathbf{0.25}$$

A12

$$1. P(D) = 0.10(0.4) + 0.04(0.6) = 0.040 + 0.024 = \mathbf{0.064}$$

$$2. P(M_A | D) = 0.040/0.064 = \mathbf{0.625}$$

A13

$$1. \text{Sensitivity} = P(+ | C) = 1 - 0.02 = \mathbf{0.98}$$

$$2. P(+) = 0.98(0.03) + 0.07(0.97) = 0.0294 + 0.0679 = 0.0973. P(C | +) = 0.0294/0.0973 \approx \mathbf{0.302}$$

$$3. P(C | -) = 0.02(0.03) / [0.02(0.03) + 0.93(0.97)] = 0.0006/0.9027 \approx \mathbf{0.000665}$$

$$4. P(\text{error}) = 0.07(0.97) + 0.02(0.03) = 0.0679 + 0.0006 \approx \mathbf{0.0685}$$

A14

$P(A \cap B) = 0.3 \times 0.6 = 0.18$. From addition rule: $P(B) = 0.72 - 0.6 + 0.18 = 0.30$. Independence test: $P(A)P(B) = 0.6 \times 0.30 = 0.18 = P(A \cap B)$. **A and B are independent** (and not mutually exclusive).

A15

$P(\text{pass}) = 0.8(0.6) + 0.3(0.4) = 0.48 + 0.12 = 0.60$. $P(\text{studied} \mid \text{passed}) = 0.48/0.60 = \mathbf{0.80}$

SOLUTIONS – SHEET 5

A16

1. Continuous RV. Range $(0, \infty)$.
2. Discrete RV. Range $\{0, 1, 2, \dots\}$.
3. Continuous RV. Range $\approx (-10, 60)^\circ\text{C}$.
4. Discrete RV. Range $\{1, 2, 3, \dots\}$.
5. Continuous RV. Range $[0, 1]$.

A17

1. $P(M = k) = (2k - 1)/36$, $k = 1, \dots, 6$. e.g. $P(M = 3) = 5/36$.
2. $P(T = k) = (6 - |k - 7|)/36$, $k = 2, \dots, 12$. Maximum at $T = 7$: prob = $1/6$.
3. $P(D = k) = (6 - |k|)/36$, $k = -5, \dots, 5$. Symmetric about 0.

A18

$\binom{10}{2} = 45$ pairs. $P(X = -20) = 15/45 = 1/3$; $P(X = 10) = 12/45 = 4/15$; $P(X = 40) = 1/45$;
 $P(X = -10) = 12/45 = 4/15$; $P(X = 20) = 4/45$; $P(X = 0) = 1/45$. Sum = 1. ✓

A19

$X = 2 + C - K$ where $C \in \{1, 2, 3, 4\}$ and $K \in \{2, 3\}$ equally likely.

X	$P(X = X)$
0	$P(C = 1)P(K = 3) = 0.1 \times 0.5 = 0.05$
1	$0.1 \times 0.5 + 0.2 \times 0.5 = 0.15$
2	$0.2 \times 0.5 + 0.5 \times 0.5 = 0.35$
3	$0.5 \times 0.5 + 0.2 \times 0.5 = 0.35$
4	$0.2 \times 0.5 = 0.10$

SOLUTIONS – SHEET 6

A20

1. $\sum 0.25(0.75)^x = 0.25/(1 - 0.75) = 1$. All ≥ 0 . **Valid PMF.**

2. $p(0) = \frac{1}{5}(-3) = -0.6 < 0$. **Not a PMF** — negative values.

3. $p(0) = \binom{5}{0} \binom{4}{-2} / 36$. $\binom{4}{-2}$ undefined. **Not a PMF.**

A21

1. $k(1 + 1 + 1/2 + 1/6 + 1/24) = k(65/24) = 1 \Rightarrow k = 24/65 \approx \mathbf{0.3692}$

2. $P(X < 2) = p(0) + p(1) = k + k = 2k = 48/65 \approx \mathbf{0.738}$

A22

Recover PMF from jumps: $p(0) = 1/2, p(1) = 1/10, p(2) = 1/5, p(3) = 1/10, p(4) = 1/10$. Sum = 1. ✓

A23

$$P(X = 3) = 0.125, P(X = 0) = 0.375.$$

$$E(X) = 0(0.375) + 1(0.3) + 2(0.2) + 3(0.125) = \boxed{1.075}$$

$$E(X^2) = 0 + 0.3 + 0.8 + 1.125 = 2.225. V(X) = 2.225 - 1.075^2 = \boxed{1.069}$$

A24

$$E(X) = 0(0.05) + 1(0.15) + 2(0.35) + 3(0.35) + 4(0.10) = \boxed{2.30}$$

$$E(X^2) = 0.15 + 1.40 + 3.15 + 1.60 = 6.30. V(X) = 6.30 - 2.30^2 = 6.30 - 5.29 = \boxed{1.01}$$

A25

$$\text{Normalise: } k = \boxed{2}. \text{ So } f(x) = \frac{3}{2}x - \frac{3}{4}x^2 \text{ on } [0, 2].$$

$$E(X) = \left[\frac{1}{2}x^3 - \frac{3}{16}x^4 \right]_0^2 = 4 - 3 = \boxed{1}$$

$$E(X^2) = \left[\frac{3}{8}x^4 - \frac{3}{20}x^5 \right]_0^2 = 6 - 4.8 = 1.2. V(X) = 1.2 - 1 = \boxed{0.2}$$

A26

$$X \sim \text{Bin}(5, 0.5). P(X \geq 4) = \binom{5}{4}(0.5)^5 + \binom{5}{5}(0.5)^5 = (5 + 1)/32 = \boxed{0.1875}$$

A27

$$1. P(X \leq 1) = (0.9)^2 + 2(0.1)(0.9) = 0.81 + 0.18 = \boxed{0.99}$$

$$2. P(X \leq 2) = 0.6561 + 0.2916 + 0.0486 = \boxed{0.9963}. \text{ Four-engine more reliable.}$$

A28

$$X \sim \text{Bin}(15, 0.3). P(X > 3) = 1 - P(X \leq 3).$$

$$P(X \leq 3) = 0.00475 + 0.03052 + 0.09156 + 0.17004 = 0.297. P(X > 3) = \boxed{0.703}$$

A29

1. $P(X \geq 1) = 1 - e^{-10} \approx \mathbf{0.99995}$

2. $P(X \leq 15) \approx 0.9513 \geq 0.95$. Need stock of at least **15 ice creams**.

A30

Rate: 1/500 per hour. Over 2000 hours: $\lambda t = 4$. $X \sim \text{Poi}(4)$. $P(X = 3) = e^{-4} \cdot 64/6 \approx \mathbf{0.1954}$

SOLUTIONS – SHEET 9

A31

1. Proc 1 ($\text{Bin}(10, p)$, reject if ≥ 1): $p = 1\%$: **0.096** ; $p = 10\%$: **0.651** ; $p = 30\%$: **0.972**

2. Proc 2 ($\text{Bin}(20, p)$, reject if ≥ 2): $p = 1\%$: **0.017** ; $p = 10\%$: **0.608** ; $p = 30\%$: **0.992**

3. **Procedure 1 generally preferred** — far better at detecting 10% defect batches (65% vs 61%).

A32

$X \sim \text{Poi}(8)$. Cumulate: $P(X \leq 12) = 0.9362 < 0.95$; $P(X \leq 13) = 0.9658 \geq 0.95$. Order when stock falls to **13 motors**.

A33

1. $P(X \geq 10) = (5/6)^{10} \approx \mathbf{0.1615}$

2. $P(X = 5) = \frac{1}{6}(5/6)^5 \approx \mathbf{0.0670}$

A34

$X \sim \text{NegBin}(50, 0.15)$. $E(X) = 283.3$, $\text{SD} \approx 43.5$. Standardise $z = (250 - 283.3)/43.5 \approx -0.765$.

$P(X \geq 250) \approx \Phi(0.765) \approx \mathbf{0.778}$

A35

$$N = 10, K = 3, n = 4. \binom{10}{4} = 210. P(X \geq 2) = (63 + 7)/210 = 70/210 = \frac{1}{3} \approx 0.333$$

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